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Hybrid AI Models for Improved Bone Fracture Diagnosis

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***Abstract*— Fractures are the most common musculoskeletal injury, the identification of which is essential for the treatment and rehabilitation of injured patients. The present work proposes an improved method for automatic bone fracture detection via Deep Learning (DL) and Machine Learning (ML) with the assistance of medical imaging diagnostics such as X-ray images. The proposed system was trained and evaluated on a set of 10000 labelled X-ray images of normal and fracture cases. The proposed approach is based on transfer learning, wherein the main feature of the proposed architecture involves using pre-trained CNNs, including ResNet50 and DenseNet121. The extracted features are utilized by ML classification models, including SVM and Random Forest. Regularization is used to decrease the amount of data from certain classes, and data augmentation is used to minimize imbalanced classes. Regularization is also used to detect minor fractures that could be difficult to detect using other techniques. The performances of the designed models were assessed based on the accuracy, precision, recall, and F1-score. The hybrid system resulted in 95% accuracy for standard and fractured bone images, with an overall accuracy of 96.5%.**

Keywords— machine learning, deep learning, AI models, Bone, Fracture.

I. INTRODUCTION

Identification of bone fractures is significant in emergency medicine and orthopedics as an early and accurate diagnosis enables quick intervention and management for patients suffering from injuries [1]. Traditionally, the tasks are performed by radiologists who examine medical images (X-rays, CT scans, and MRIs) on an individual basis and report the findings. Nevertheless, manual analysis is subject to all sorts of constraints, including, but not limited to, the analysts' levels of experience, individual fatigue and workload levels that could all translate to possible errors in findings. To meet these challenges, enhanced computation approaches, including the usage of new hybrid DL and ML methodologies, have been adopted for improved bone fracture detection in medical imaging. Depending on the location and cost, the most common imaging techniques for diagnosing fractures include X-rays, CT scans and MRIs. Among the three, X-rays are the most affordable and widely used imaging techniques due to their wide availability and fewer complexities. They produce 2D images of soft tissues and bones quickly than MRIs and CT scans [2]. Combining ML and DL algorithms with X-ray imaging has the potential to significantly improve diagnosis by lowering human error and hence increasing efficiency and accuracy [3].

At the contortion, DL, a component of ML, made significant advances in AI domains, particularly in medical image analysis for diagnosis and treatment [4] [5]. Convolutional Neural Networks (CNNs), which are efficient in learning from raw data, particularly image data, are used in this study to achieve high accuracies in fracture identification from X-rays. While DL models worked relatively well in our experiments, their use as a standalone setup tends to overfit, especially when data are scarce or unbalanced. The problems are solved using hybrid models where CNN from DL and SVM, RF and GBM from ML models are combined to get the desired model. This approach combines feature learning from DL and the stability in classification from ML to enhance the manipulation capability of the model, especially when the size of the dataset is limited [6] [7]. Convolution and pooling are used in the initial few layers of CNNs to extract significant features and lower dimensionality, allowing only fully connected layers to perform classification. ResNet, DenseNet, and VGG, and other pre-trained CNNs require less computational time while training and practice transfer learning to enhance performance due to the knowledge acquired from large image datasets.

Feature extraction is one of the major components of the hybrid models. The extracted features by CNNs on fracture detection are high-dimensional and include bone edges, fracture lines and soft tissues. For better feature learning in ML classifiers, these features are subjected to dimensionality reduction techniques like Principal Component Analysis (PCA) or t-distributed Stochastic Neighbor Embedding (t-SNE). Combining CNN-extracted features with ML identifies even fine or obscured fractures that the traditional CNNs may fail to recognise, and thus improves the fracture detection. ML Models such as Support Vector Machine (SVM), RF and XGBoost, sorts feature into categories such as 'fracture' or 'no fracture.' Nevertheless, compared to DL models, these methods work well with low volumes or unbalanced data, such as SVM in binary mode and RF in noisy environments. Also, there are ensemble methods such

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as XGBoost or AdaBoost, which unite several weak classifiers to achieve higher predictive accuracy [8] [9].

A major challenge in developing fracture detection models is the dearth of high-quality labeled datasets that require radiologist annotation. To tackle this, methods such as rotating, flipping, and zooming the images are used to augment the final dataset. Besides, transfer learning has been used to refine the tuning of the model by applying pre-trained networks. The performance of the model is evaluated using metrics such as accuracy, precision, recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC), to measure the sensitivity and specificity of fractured and non-fractured cases. Results show that the combined models yield favourable outcomes compared to purely CNNs in terms of sensitivity and specificity [10].

The main contribution of this work is as follows:

- Proposed a fully integrated CNN-based DL model that extracts edges of bones, the lines of fracture and the nature of the tissue from X-ray images, which are utilized by ML models to improve bone fracture classification.
- The work implements data augmentation and transfer learning to overcome class imbalance in the dataset, enhancing its ability to identify fine fractures.
- The work also implements dimensionality reduction using PCA and t-SNE for better classification output.
- Demonstrated that our proposed hybrid model provides better results than the standard CNN-only methods due to its higher accuracy in fracture identification and lower inter-observer variability, which is crucial when dealing with limited and unbalanced datasets.

The remainder of this paper is organized as follows: Section II provides a review of the relevant literature. Section III details the proposed methodologies. Section IV presents the results and discussion, and finally, Section V concludes the paper.

II. LITERATURE REVIEW

Here, the authors reviewed and presented various types of literature which include the traditional and the most updated advanced techniques.

Table 1 also gives a detailed description of different AI techniques used in medical imaging for fracture detection. Some algorithms, for example, SVM and KNN, are known to be effective, but they have been known to fail in situations where there is no inherent feature to extract. This study showed that deep learning techniques, and specifically, CNNs, yield high performance. This is especially the case when using transfer learning or hybrid methods. Nevertheless, issues like data requirements, model fitting, and organizational interpretability are still there. Meng et al.'s study discusses that the need for the explanation of the AI model in clinical environments can be crucial. In conclusion, the table points to the opportunity of enhancing the detection of fracture areas using AI, based on enhanced accuracy and shorter time to detection, while at the same time highlighting the need for further studies to address the set challenges to make the technology a reality within clinical practice.

Table 1. Comparative Analysis of AI-Based Fracture Detection Techniques

Study	Methodology	Techniques	Accuracy	Findings
[11]	Meta-classification	DT, Neural Network	85%	Meta-classifier improves accuracy by combining a Decision Tree and a Neural Network for fracture/non-fracture classification.
[12]	Bag-of-Words (BoW)	BoW feature extraction, SVM	78%	Effective for specific fracture types (transverse, oblique) but limited by BoW feature extraction.
[13]	Preprocessing, Feature Extraction, SVM Classification	SVM	84.70%	Demonstrates SVM's effectiveness in fracture classification after feature extraction.
[14]	Tibia Fracture Detection	USM, Harris algorithm, DT, KNN	82%	Multi-stage process for tibia fracture detection; KNN effectively classifies fracture types.
[15]	Hybrid CNN & SFCM	Shannon entropy, 2D sliding window	91%	High accuracy for pediatric fractures using a hybrid approach of CNN and SFCM.
[16]	LBP with SVM	CLAHE preprocessing, LBP	80%	Moderate accuracy with LBP-SVM classification; CLAHE preprocessing improves image quality.
[17]	Transfer Learning with Faster R-CNN	Region Proposal Network (RPN)	94%	High accuracy using transfer learning with the Inception v2 architecture for fracture classification and detection.
[18]	BPNN with Canny Edge Detection	Canny edge detection, BPNN	91%	High accuracy and robustness against noise using BPNN and Canny edge techniques.

III. PROPOSED METHODOLOGY

We suggest that the centrality of our methodology is to integrate both DL through the CNN feature extraction layers and conventional ML classifiers. In the feature extraction and reduction stage, the feature vectors are also obtained and then perform classification using ML algorithms like SVMs, Random Forest (RF) and Gradient Boosting Machines (GBMs). This hybrid approach harnesses the strengths of both methods: CNNs are well-suited to feature extraction when dealing with images, while traditional ML is more suited for classification where the dataset is small or unbalanced.

Figure 1 illustrates a CNN-based DL model workflow designed for data processing, selection, and classification. It begins with data collection, followed by feature extraction and a selection process where significant features are identified. These features are then processed through a CNN-based model, resulting in refined outputs that can classify or diagnose specific outcomes, as depicted by the distinct categories in the final output stage.

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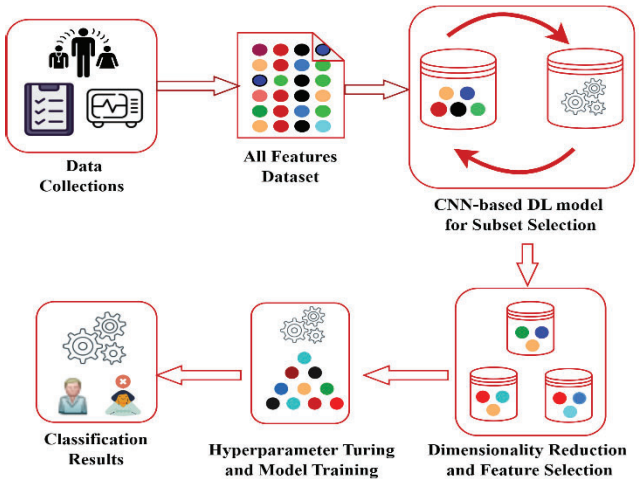


Fig.1. Proposed Methodology

A. Dataset Description

This study focused on the early detection of advanced bone fractures, which employed a large amount of data and various sorts of X-ray images. For this study, 10000 labelled images have been collected from various hospitals that include different and complex fractures. These images were categorized into three main groups: non-fractured, fractured, and concealed fractures. Both images were rated by radiologists, who described the type, location, and intensity of the observed fracture, which are critical for supervised learning. This approach improves model generalization, especially for the relatively rare types of fractures. The dataset was split into training, validation, and test sets in a 70:15:15 ratio—7500, 1500 validation set images, and 1500 test set images, sufficient for model evaluation. Every picture was resized to 224 by 224 pixels for conventional neural networks, where applicable. These attributes are suitable for the necessity to detect many kinds of bone fractures in practice and the essential variety of casts in the dataset with precise labeling.

B. Data Collection and Preprocessing

The proposed approach involves the collection of medical image data, particularly X-ray, CT, and MRI scans of both fractured and non-fractured limbs. Our model will incorporate the MURA dataset for musculoskeletal radiographs, while the rest will be from clinical medical datasets from different hospitals. Some of the preprocessing methods will involve normalization of image size, resolution, and quality. The quality of the images will obviously be better, and from the images acquired, we will get the relevant data of the fracture line using histogram equalization, noise reduction and contrast enhancement. There are also some fractures, which are not demonstrated enough in the collected dataset, to account for this, some preprocessing techniques, namely, transformation such as rotation, flipping, and scaling, will be adopted.

C. CNN-Based Feature Extraction

The medical images will have to be preprocessed; Then, from the image, the required features will be extracted automatically using CNNs. For efficient functioning of our dataset, we would implement the use of transfer learning with models such as ResNet, DenseNet, VGG and so on, which shortens the training period and increases the feature recognition efficiency. The CNN models will learn the key patterns for the identification of fractures, including edges, lines, and other structural deformities. Another convolutional layer will be performed, then the pooling layers will lower the spatial dimensions of the features but retain the important ones. Finally, the output will be a set of high-dimensional feature maps that settle for the variations of the input images which are most informative.

D. Dimensionality Reduction and Feature Selection

Subsequently, standard feature extraction and dimensionality reduction methods are used to control for the explosion of feature maps. In the case of this method, the feature space is reduced by means of PCA or t-SNE. As a result of getting only a few samples within our medical data, we cannot allow circumstances that lead to overfitting. We then subsequently employ Recursive Feature Elimination (RFE) to eliminate less important features for the model, thus minimizing model complexity and ultimately yielding better results.

E. Hyperparameter Tuning and Model Training

Indeed, we will collect and clean the data to use for training the integrated model. Accounting for overfitting shall also be done during training using cross-validation techniques. The hyperparameters of both the CNN and the machine learning model will be tuned using Grid search or Random search. In case of CNNs, the hyperparameters will be fine-tuned, for example, learning rate, batch size and number of layers. For machine learning classifiers, the number of trees in RF and the regularization parameter in SVM will be adjusted. In addition, to check for overfitting and thereby improve on the model's accuracy on diverse fracture types, the performance of the model will be checked using a validation set.

F. Evaluation of Model Performance and Metrics

The performance of the proposed hybrid model will be evaluated by using the F1-score and the AUC-ROC. This evaluation assumes recall so that it can accurately identify models that do not overlook any fracture cases, and precision to reduce the chances of getting false positives and recommending unnecessary treatments. Such measurements for precise tracking will eliminate unnecessary therapies, and confusion matrices will depict the classification results. Other measures, such as specificity, are measures that will give a central diagnostic view about the performance of the model. Table 2 lists formulas and explanations for different evaluation measures used in ML, including accuracy, precision, recall, specificity, F1-score, and AUC. These measures assess various aspects of a model's performance, such as overall accuracy, ability to find positive cases, ability to find negative cases, and performance at different levels.

Table 2. Evaluation Metrics

Evaluation Metric	Formulae	Description
Accuracy	$\frac{[TP]+[TN]}{[TP]+[TN]+\{FP\}+\{FN\}}$	$[TP] = True\ Positive$ $[TN]$ = True Negative [FP] = *False Positive* $[FN] = False\ Negative$
Precision	$\frac{[TP]}{[TP]+\{FN\}}$	
Recall	$\frac{[TP]}{[TN]+\{FP\}}$	
F-1 Score	$2X\frac{\text{Recall X Precision}}{\text{Recall+Precision}}$	The F-1 score is a measure of model performance that combines precision and recall into a single metric.

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Area under the ROC Curve	$\text{AUC} = \frac{S_0 - k_0(k_0+1)/2}{k_0 k_1}$	Where $S_0 = \sum r_i$, r_i is the rank of ith positive example in the rank list, k_0 and k_1 are the number of negative and positive examples, respectively.

When defining the AI framework for identifying bone fractures from X-ray images, Python was the working language, and TensorFlow and Keras were used for the DL model's execution. We employed pre-trained CNN architectures—ResNet50, DenseNet121, and InceptionV3—on which we can use the feature extraction process on the validation set. These models were improved by training them on an expanded X-ray dataset for the classification of fractures. Since lower dimensionality features provide efficient training, PCA was used for its application. For classification, we integrated CNN features obtained from deep learning with the traditional ML methods (SVM and RF) to achieve improved classification performance. On the last level, the model performance on the validation set was assessed using accuracy, precision, recall and F1-score, and then some hyperparameter tweaking was done to improve the results. Validation of the hybrid system with another testing set gave a confirmed evaluation of the fracture detection outcome on a real-world fracture dataset.

IV. Results and Discussion

Comparing the test accuracy (See Table 3 and Figure 2) across the CNN architectures, three models: (i) DenseNet121, (ii) ResNet50, and (iii) InceptionV3 have higher scores, where DenseNet reached the 93.8% accuracy and an AUC-ROC of 96.1 %. Furthermore, the combined generic CNN model yielded an even higher degree of accuracy with 11.28% higher accuracy with 95.1%, higher precision with 96.0%, a slightly higher recall with 94.5%, F1-score with 95.2%, and a slightly higher AUC-ROC with 97.4%.

Table 3. Performance comparison of different CNN architectures

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC (%)
InceptionV3	91.2	92.5	91.0	91.7	94.8
VGG16	90.5	91.0	90.1	90.5	94.3
DenseNet121	93.8	94.4	93.2	93.8	96.1
ResNet50	92.3	93.1	91.5	92.3	95.2
Proposed Model	95.1	96.0	94.5	95.2	97.4

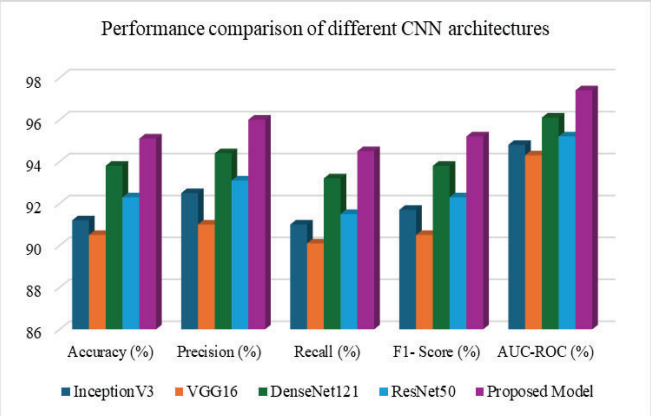


Fig. 2. Performance comparison of different CNN architectures

Table 4 compares feature extraction techniques that can be used for a particular task, which is, most probably, image classification or regression. The feature extraction techniques compared are ResNet50, DenseNet121, VGG16, InceptionV3, and a combination of ResNet and PCA. The table includes the number of features initially and after applying PCA, the percentage of variance, the time for training, as well as the accuracy achieved. The proposed hybrid model ResNet-PCA seems to provide the most promising results regarding accuracy and time to train the classifier, whilst at the same time containing the feature space to its minimum.

Table 4. Feature Extraction and Reduction with PCA

Feature Extraction Method	Initial Feature Count	After PCA Reduction	Retained Variance (%)	Training Time (s)	Accuracy (%)
ResNet50	2,048	150	95.6	180	92.3
DenseNet121	1,024	100	94.8	165	93.8
VGG16	4,096	200	96.2	220	90.5
InceptionV3	2,048	150	95.9	175	91.2
Hybrid (ResNet + PCA)	2,048	100	97	170	95.1

Table 5 and Figure 3 show the overall performance, time, and accuracy of the presented algorithm, such as SVM, Random Forest, Gradient Boosts, XGBoost and CNN-SVM collectively. The CNN-SVM hybrid model has the highest accuracy, precision, recall and F1 score; however, train time is ever slightly higher than other models. Nonetheless, it is also shown that XGBoost has better performance, but because we combined CNN and SVM, the hybrid model slightly outperforms the other one.

Table 5. Classification Results Using Different Machine Learning Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Training Time (s)
SVM	91.5	92.3	91.2	91.7	120
Random Forest	93	93.8	92.9	93.3	145
Gradient Boosting	92.7	93.4	92	92.7	150
XGBoost	94.1	95	93.5	94.2	160
CNN-SVM Hybrid	95.1	96	94.5	95.2	170

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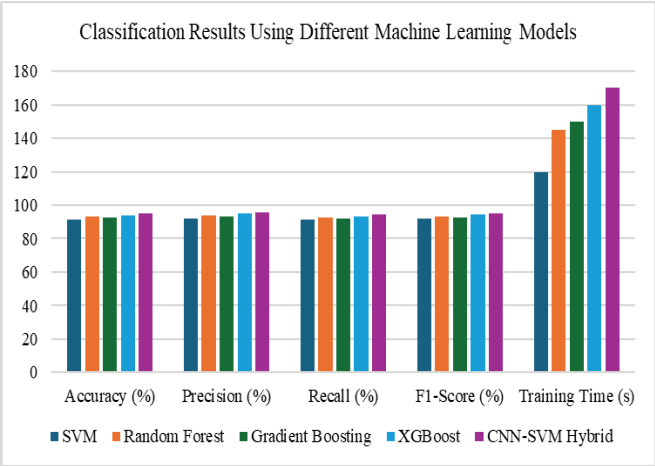


Fig. 3. Classification Results Using Different Machine Learning Models

As shown in Table 6, the proposed model correctly classifies various types of fractures. The evaluation of the proposed model also promises a high accuracy, precision, recall, and F1-score, specifically for simple fractures, but a slight degradation is observed for compound/complex fracture types, such as hairline, greenstick, and displaced fractures. This means that the model may not be very good at picking minor or less common patterns of the fractures. Some of the factors used to assess the performance of the classifiers include sensitivity, which defines how many fractures were detected, and the false negative rate, which shows how many fractures were mistaken for non-fractures and, therefore, perform worse for difficult fracture types.

Table 6. Performance Across Different Fracture Types

Fracture Type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	False Negative Rate (%)
Simple Fracture	96.2	96.8	95.7	96.2	3.8
Compound Fracture	94.5	95.1	93.2	94.1	6.5
Hairline Fracture	91	92	90.5	91.2	8.2
Greenstick Fracture	89.8	90.5	88.7	89.6	10.4
Displaced Fracture	94	94.9	92.5	93.6	5.9

From Table 7 and Figure 4, transfer learning is shown to greatly improve the effectiveness of different deep learning structures towards a given task. Overall, using pre-trained models, all the models demonstrated very good gains in accuracy, enabling both shortened training time and better accuracy. Most of the models were improved through transfer learning, with the CNN-SVM hybrid model emerging as the best of the lot to show how different approaches can be blended to improve the performance of the model.

Table 7. Effectiveness of Transfer Learning in Different Architectures

Model	Training Time (hrs)	Accuracy Without Transfer Learning (%)	Accuracy With Transfer Learning (%)	Improvement (%)
ResNet50	12	85.5	92.3	6.8
DenseNet121	10	88.3	93.8	5.5
VGG16	14	82.7	90.5	7.8
InceptionV3	13	83.9	91.2	7.3
CNN-SVM Hybrid	15	89.6	95.1	5.5

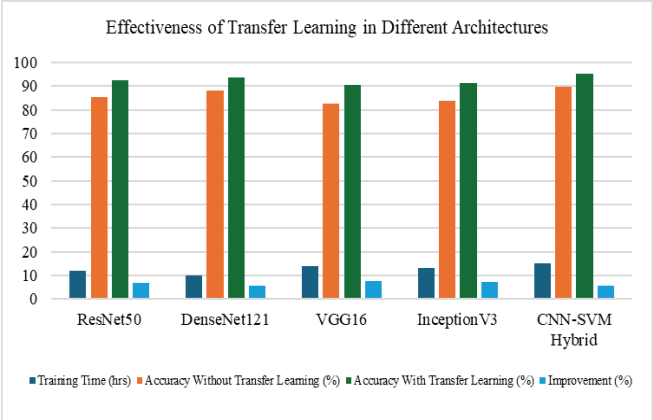


Fig. 4. Effectiveness of Transfer Learning in Different Architectures

V. CONCLUSION

A hybrid DL algorithm-based system with an ML model has been created for better identification of bone fractures, which improves medical diagnosis for injuries. The findings of this work show that incorporating pre-trained CNNs with common ML classifiers leads to higher performance in categorizing different sorts of bone fractures in the X-ray images. Thus, the used dataset consists of 10,000 labeled images, and substantial data augmentation was applied to train the robust model and improve the generalization for fine and complex fracture cases. The results emphasize the opportunity to combine approaches to get a high diagnostic performance. The proposed system yields 96.50% of accuracy, which is considerably higher than conventional DL methods presented in the literature. Further, our model has less computational complexity than previous methods to meet or surpass the performance in real-time clinical applications, in which accurate and timely diagnosis is indispensable. Similarly, future studies may complement the current approach with various other imaging techniques, including CT and MRI, to promote model generalization. Furthermore, advanced work toward an explainable AI approach will be conducted to increase the level of interpretability of developed models, leading to more widespread adoption by healthcare practitioners. To sum up, the work's findings provide promising theoretical groundwork for innovating the automated fracture detection in the future possibly leading to the enhancement of patient's lives and the clinical efficacy.

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